

ML Practice Projects — Roadmap

Consolidation phase after Geron Chapters 2-4 · 4 Classification + 3 Regression · updated 2026-06-26

Purpose

After finishing Geron Chapters 2-4, the goal is to consolidate by building practice projects before moving deeper into the book. Each project is a permanent **reference notebook** and must add a NEW skill beyond the previous one. Classification and regression are interleaved so the evaluation concepts keep contrasting.

Completed

P1 - UCI Heart Disease (Cleveland) · binary · **DONE**

Path: classification/project_4_UCI_Heart_Disease/Heart_Disease_P1_Clinical_Prediction.ipynb

Template covering full EDA (skew, outliers, chi-square, ANOVA, VIF, Spearman + effect sizes), custom **AgeGroupImputer** (grouped imputation), ColumnTransformer pipeline, 12-model bake-off, GridSearch tuning, and a **Rigor & Honesty** section (demographics-only check, RepeatedStratifiedKFold distribution, error bars, Youden-J threshold, learning + validation curves). Ends with a QUICK-RECALL cheat-sheet.

The 7-Project Roadmap

#	Project	Type	New skills beyond P1
1	Telco Customer Churn	Binary (imbalanced)	SMOTE / class_weight, PR-AUC, calibration, cost-based threshold, high-cardinality encoding
2	Ames Housing	Regression	80 features, ordinal encoding, heavy missingness, log-transform target, feature selection
3	Fetal Health (CTG)	Multiclass (medical)	macro-F1, K x K confusion matrix, one-vs-rest ROC, multiclass + imbalance together
4	Medical Insurance Cost	Regression	interaction terms (smoker x age x BMI), small clean data, interpretation focus
5	Covertypes	Multiclass (big data)	scale (581k rows), sampling, efficiency / training-time tradeoffs
6	Energy Efficiency	Regression	multi-output (heating + cooling load), tiny dataset, near-deterministic signal
7	Glioma Grading	Binary (medical / genomics)	clinical + gene-mutation (binary) features, clinical-only vs +molecular contribution, biological interpretation (IDH1 etc.)

Suggested Order (interleaved)

Churn → Ames → CTG → Insurance → Covertypes → Energy → Glioma

Difficulty ramps gently: Churn and Insurance reuse the P1 pipeline; Ames (feature breadth) and Covertypes (data scale) are the heavy ones. Glioma is the genomics capstone, done last.

Project 7 references

Dataset: UCI #759 - Glioma Grading Clinical and Mutation Features (839 patients, ~23 features, target = grade LGG vs GBM).

archive.ics.uci.edu/dataset/759/glioma+grading+clinical+and+mutation+features+dataset

Reference paper: MDPI Int. J. Mol. Sci. 2022, 23(22), 14155 - mdpi.com/1422-0067/23/22/14155

The 5 rules that never change: (1) split before you learn anything; (2) beat the dumb baseline; (3) right metric, not just accuracy; (4) significance is not strength - pair p-values with effect size; (5) distrust scores that are too good.

Next up: Telco Customer Churn

Data: Kaggle **blastchar/telco-customer-churn** (~7,000 rows, ~27% churn). New muscle: the imbalance toolkit - PR-AUC over ROC, SMOTE vs class_weight, calibration curves, cost-based threshold.

Known gotcha (catch it in EDA): TotalCharges is stored as text with blank strings for tenure-0 customers.